THE KOSZUL COMPLEX OF A PRODUCT IS THE CONE OF A MAP BETWEEN TWO KOSZUL COMPLEXES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The Koszul complex of an extension by an element is the cone of multiplication by that element. Lemma \ref{lemma-cone-koszul-abstract}, cited at proof lines 6859-6861. Identifies the three complexes with cones.
- The abstract comparison of the cone for a product with a map between the cones for the factors. Lemma \ref{lemma-cone-squared}, cited at proof lines 6862-6864. Gives the homotopy equivalence claimed.

Source reference. Original lemma 062B: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7790–7790. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0628: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7787–7787. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 062A: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7789–7789. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring. Let $\varphi : E \to R$ be an R-module map. Let $f, g \in R$. Set $E' = E \oplus R$ and define $\varphi'_f, \varphi'_g, \varphi'_{fg} : E' \to R$ by φ on E and multiplication by*

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f, g, fg on R. The complex $K_\bullet(\varphi'_{fg})$ is homotopy equivalent to the cone of a map of complexes

$$K_\bullet(\varphi'_f)[1] \longrightarrow K_\bullet(\varphi'_g).$$

Proof. By Lemma 0628 the complex $K_\bullet(\varphi'_f)$ is isomorphic to the cone of multiplication by f on $K_\bullet(\varphi)$ and similarly for the other two cases. Hence the lemma follows from Lemma 062A. □